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TECHNICAL REPORT

Emerging Contaminants

Per- and polyfluoroalkyl substances in agricultural systems: Human dietary exposure via beef raised on corn stover from a field with contaminated biosolids

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Abstract

Per- and polyfluoroalkyl substances (PFAS) in biosolid-amended soils can transfer and accumulate in crops, cattle, and people. Bioaccumulation factors (BAFs) are often applied to estimate the transfer of contaminants from soil to crops. However, they can vary widely and introduce uncertainty to exposure and risk estimates. We, therefore, aimed to quantify this uncertainty in a case study of an agricultural field with elevated soil concentrations of perfluorooctane sulfonic acid (PFOS) using literature-derived BAF versus measured concentrations of PFOS in corn (*Zea mays* L.) kernels and stover. PFOS was the predominant PFAS detected in soil and corn stover (<100 and 19 ng/g), and no detectable PFAS were identified in kernels. The median BAF (0.17) for PFOS was similar to that derived from a review of previous studies, while the maximum (0.2) was over an order of magnitude lower. Median PFOS concentrations in stover from our samples were comparable (16.60 ng/g) to those calculated using the literature-based BAF (16.28 ng/g). For cattle consuming stover, median and upper bound concentrations of PFOS in beef (30 ng/g) were similar and 60% lower using measured versus literature-derived BAF concentrations in stover. Finally, the central tendency exposure for children (27 ng/kg-bw/day) was similar using measured versus literature-derived BAF concentrations in stover and

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higher compared to adults (15 ng/kg-bw/day). Overall, these results indicate that (1) corn kernels accumulate little to no PFAS even when soil concentrations are elevated, (2) direct measurement of PFAS in crops can reduce uncertainty in exposure and risk assessment, and (3) PFOS can biomagnify via the soil-stover-cattle-human pathway and is found to pose a potential risk in our case study.

Plain Language Summary

Per- and polyfluoroalkyl substances (PFAS) are long-lasting chemicals used in many products that can build up in soil, water, crops, and animals. This study looked at an agriculture field where sewage sludge (biosolids) containing PFAS had been used for over a decade. We measured PFAS in the soil and corn plants and used computer tools to estimate how much could end up in cattle and people. While PFAS levels in soil and corn stover (leaves and stalks) were high, no PFAS were found in corn kernels. We used real measurements and published values from other studies to estimate how much PFAS could be found in beef and liver from cattle eating the contaminated corn. The results showed that even moderate PFAS levels in soil and feed can lead to high amounts in beef and liver—above safety limits set by US and European health agencies. Children were especially at risk due to eating more food for their body size. Our study shows why testing real crops is important to protect food safety.

1 | INTRODUCTION

Per- and polyfluoroalkyl substances (PFAS) have been used historically and have bioaccumulation potential (Kwiatkowski et al., 2020). Numerous studies have shown that exposure to some PFAS can adversely affect multiple systems in the body, including the immune, reproductive, and endocrine systems, as well as liver, kidney, and thyroid function (Fenton et al., 2021). While drinking water containing PFAS is an important exposure pathway, diet is an important pathway for the general population (Seshasayee et al., 2021). Uptake and accumulation of PFAS into foods can occur through contaminated wastewater, soil, and biosolids, especially in farming communities that rely on local food sources (Bolan et al., 2021; Brendel et al., 2018; Costello & Lee, 2020; De Silva et al., 2021; EFSA Panel on Contaminants in the Food Chain [EFSA CONTAM Panel] et al., 2020; Evich et al., 2022).

Wastewater treatment plants (WWTPs) have been identified as sources of PFAS to the environment, and elevated concentrations have been traced to industrial effluent (e.g., metal plating facilities) and landfill leachate (Ellison, 2022). Conventional WWTP methods, such as activated sludge and chemical coagulation, may not substantially reduce PFAS and can even lead to the transformation of polyfluoroalkyl precursors into terminal products, such as perfluorooctanesulfonic acid (PFOS) (Gagliano et al., 2020; Lenka et al., 2021). Elevated concentrations of PFAS have been identified in WWTP influents and effluents as well as sludge, the solid waste component that can be land-applied to agricultural fields as biosolids to improve soil health and crop production (Fu et al., 2023). PFAS can accumulate in soil with repeated application of biosolids, are taken up into plants, and accumulate in animals (B. A. Astmann et al., 2025; Simones et al., 2024). This accumulation pathway can be especially amplified for integrated crop-livestock systems (Death et al., 2021).

PFAS uptake in plants occurs via diffusion and sorption through the roots and can translocate to the shoot and fruit (Jha et al., 2021; W. Wang et al., 2020). Field and greenhouse studies have reported that grasses, leaves, and roots typically accumulate more PFAS than kernels and fruits of cereal crops (Brown et al., 2020; Ghisi et al., 2019; Lee et al., 2020; Xu et al., 2022). Cattle and other livestock consuming food grown in affected soils can accumulate elevated concentrations of PFAS in their muscle and milk, posing a potential risk for human consumption (Drew et al., 2021; Simones et al., 2024).

As direct measurement of PFAS in animal products is not always possible, fate and transport models are often used to estimate concentrations in environmental media and foods. Uptake into plants from soil is often represented in these models as bioaccumulation factors (BAFs) (Xu et al., 2022), with larger values indicating greater transfer (Bilal et al., 2022; Sunyer-Caldú et al., 2022). Longer chain PFAS, such as PFOS, often have lower BAFs in plants (<0.2) compared to shorter chain PFAS, such as perfluorobutanoic acid (PFBA; >1.5) (Navarro et al., 2017). Greater uptake has been found for short-chain PFAS with higher solubility, which enables

easier root uptake and xylem translocation (Battisti et al., 2024; Simones et al., 2024; W. Wang et al., 2020). Additionally, leaves and stalks have higher BAFs, such as PFBA (>10) and PFOS (>0.2), as the transpiration rates exceed those of the grain or fruit portions of the plant (Blaine et al., 2014; Xu et al., 2022). Growing conditions and other environmental factors can also influence uptake, as can soil properties. For example, increasing soil organic matter content has been shown to decrease PFAS uptake in lettuce (*Lactuca sativa* L.) (Li et al., 2024; W. Wang et al., 2020). The inverse has been observed for animals, with generally higher BAFs for PFOS and other longer chain PFAS compared to shorter chain PFAS (Lupton et al., 2022). However, this may vary by species; for example, in cattle, two longer chain C8 PFAS accumulate differently with a high BAF for PFOS (2–5) and a low BAF for PFOA (<1) (Drew et al., 2021; Kowalczyk et al., 2013). BAFs can differ by organ, for example PFOS accumulation is greatest in the cattle liver with concentrations in muscle tissue approximately two to three times lower (G. Wang et al., 2017). Such differences may be influenced by solubility as PFAS that are more water soluble (e.g., carboxylic acids) have more excretion via urine compared to those with lower solubility (e.g., sulfonic acid), which may rely more on biliary excretion, causing longer retention in the blood, liver, and kidney (Lupton et al., 2025).

The wide variation in literature-reported BAFs for crops and livestock may introduce considerable uncertainty in predicting concentrations in foods. Understanding and reducing this uncertainty is crucial for better managing and mitigating human exposure to PFAS through the biosolids pathway. Therefore, we aimed to quantify this uncertainty in a case study of PFAS-contaminated agricultural soil using literature-derived BAF and measured concentrations in corn kernels and stover as inputs to models estimating concentrations in beef as well as human exposure and risk.

Core Ideas

- Per- and polyfluoroalkyl substances were not detected in corn kernels, even in soils with >100 ng/g perfluorooctanesulfonic acid (PFOS) from biosolid applications.
- Measured and literature-based PFOS in stover yielded similar predicted PFOS levels in cattle tissues.
- PFOS concentrations in beef and liver exceeded EPA and EU health-based safety thresholds.
- Direct crop measurements reduced uncertainty compared to using literature-derived BAF estimates.
- Modeled PFOS exposure from beef and liver exceeded EPA's reference dose, with highest risk to young children.

2 | MATERIALS AND METHODS

2.1 | Sample collection and processing

For over a decade, a US Midwest field had repeated biosolid application from a WWTP containing elevated concentrations of PFOS (Bogdan et al., 2021). We do not have details on the biosolid application methods used. In 2017, PFOS was identified in biosolids from the same US Midwest WWTP at concentrations up to 2100 ng/g-ww, orders of magnitude above biosolids at other WWTPs and soils ranging up to 500 ng/g-dw (Bogdan et al., 2021). The field was typically planted in corn, soybean (*Glycine max* L.), and alfalfa (*Medicago sativa* L.) rotation. In 2018, field corn was planted and harvested in this agricultural field because it was anticipated to take up less PFOS than the other crops. It is a high-production crop in the US Midwest and is used as cattle feed (Costello & Lee, 2020; Jha et al., 2021; Krippner et al., 2014, 2015; Navarro et al., 2017; Stahl et al., 2009). Soil and corn plants were collected in October 2018 using an incremental sampling approach (ITRC, 2020). Field and lab staff followed state department and Interstate Technology and Regulatory Council guidelines (Bogdan et al., 2021). Equipment (polypropylene bins, stainless steel tools, and sieves) was rinsed with Alconox, water, and methanol between samples, and nitrile gloves were changed after each sample (Bodgan et al., 2021; Kidd et al., 2022). One acre (0.40 ha) was selected, and 10 sampling plots were created (Figure S1) (Bogdan et al., 2021). One corn plant with at least one ear was collected from each plot and sealed in a low-density polyethylene PFAS-free bag. While seven corn plants were cut two inches above the ground per protocol, the other three were pulled from the ground and later discarded as they were contaminated by soil on the root ball. Each soil sample (6- to 8-in. depth) was collected using a push probe next to the stalk and sealed in a 4-oz high-density polyethylene jar. Soil sampling was limited to the top 6 in., as this is typically the depth with the highest PFOS concentration from biosolids application (Sepulvado et al., 2011) and is the depth at which seventy to 80% of corn roots are concentrated (Kranz et al., 2008). Samples were processed in the lab. The cob, husk, kernels, stalk, and leaves were separated, air-dried, and weighed for each corn plant. Approximately 15 g aliquots were sealed in polyethylene zip bags and stored at −20°C. Kernel composites were created by combining equal mass (100 g each) from each plant to make three field composites (C1, C2, and C3). Each soil sample was freeze-dried, sieved, weighed, sealed in a polyethylene zip bag, and stored at −20°C.

2.2 | PFAS analysis

Samples were transported on ice for PFAS analysis and stored at −20°C. Composites of soils (n = 3) and corn kernels (n = 3) were quantified for 45 PFAS in 2020 using liquid chromatography–quadrupole time-of-flight mass spectrometry and liquid chromatography-tandem mass spectrometry (LC-MS-MS), respectively, as described by Nickerson et al. (2020) (soil) and Meng et al. (2022) (kernels) (Supporting Information Methods A and B). Reporting limits (RL), equivalent to the limit of quantification, ranged from 0.01 to 3.6 ng/g for soils and 0.025 to 0.1 ng/g for kernels, and acronyms and RLs for each PFAS are provided (Tables S1 and S2).

Soil and stover (kernel, husk, cob, stalk, and leaves) from each plot (n = 7) were analyzed by Eurofins Environment Test America in 2024 using a modified EPA Method 537 procedure. Before the PFAS analysis, each plant sample was homogenized by grinding it at room temperature using a 3.5-quart Robot Coupe Blixer 3 Series D. Samples were analyzed for 68 target analytes using a modified EPA Method 537 (US EPA, 2018, 2024) protocol. Briefly, samples were fortified with isotopically labeled extraction standards and extracted using ultrasonic extraction in methanol. The extracts were then vortexed and centrifuged. Additional extract clean-up to remove matrix interferences was not performed. A portion of the supernatant (2.0 mL) was transferred and concentrated using nitrogen in a heated water bath, then reconstituted with methanol to 1.0 mL and analyzed for target PFAS, along with isotopically labeled PFAS injection standards by LC-MS-MS. Reporting limits ranged from 0.19 to 68 ng/g-ww (Table S3). For all PFAS analyses, adjustments were made to reporting limits to account for interferences and isotope dilution techniques were used, which can help mitigate data quality issues. Laboratory control samples and sample duplicates were performed during the quality control process (Table S4).

2.3 | Data analysis

Any individual plant or field composite sample value below the RL was substituted with the RL divided by 2 for our calculations. PFAS geometric means (GM) and geometric standard deviations were calculated for each set of composites (n = 3), with individual plot results reported separately.

As described below, we conducted a detailed review of the available scientific literature to derive BAFs. Then, we used it to estimate PFAS concentrations in corn kernels and stover. Using a pharmacokinetic model, these literature-derived BAF results and measured concentrations from our samples were used to produce separate estimates of PFOS concentrations in cattle tissues (meat, liver, kidney, and serum). We then estimated PFOS exposure and risk for children and adults from consuming corn kernels and cattle meat using these two approaches (literature-derived BAF and measured). Data were analyzed using Microsoft Excel (Version 16.x) and R (Version 4.4.2), and the cattle model was validated using the ConTrans model from the German Federal Institute for Risk Assessment (based on Mikkonen et al., 2023).

2.3.1 | Literature-derived BAFs for corn kernel and stover

We conducted a preliminary meta-analysis of the transfer of PFAS from soil to corn plants (*Zea mays*) by searching scientific databases for peer-reviewed publications (PubMed, Scopus, and Web of Science). However, a formal meta-analysis could not be performed due to the limited number of available studies. Keywords used in the search included "perfluoroalkyl substances" or "perfluorinated substances" in combination with and/or "stover," "uptake," "biosolids," "maize," and "plants." References of the retrieved literature were also examined to identify additional studies. Nine publications were identified (Blaine et al., 2013; Felizeter et al., 2021; Krippner et al., 2015; Lan et al., 2020; Liu et al., 2017, 2019; Stahl et al., 2009; Wen et al., 2016; H. Zhang et al., 2018). These included three studies that added known concentrations of PFAS to soils and six of corn grown in soils near a point source of PFAS, including industrial manufacturing sites (five), landfills (two), or WWTP (two). We recorded this and other relevant information from each study, including paired concentrations of PFAS in soil and corn kernels/stover, if provided, or the reported BAFs. If results were not reported specifically as corn stover, we recorded results for individual plant parts (husk, cob, stalk, and leaves). BAFs were calculated by dividing the concentration in each of the following components (corn kernels, stover, or plant parts) by the concentration in the soil. For this case study, we focused on soils with PFOS concentrations between 10 and 1000 ng/g, as these levels were most closely aligned with our soil concentrations. We did not use the non-detect values from the literature-derived BAF approach due to the large variability in RLs across studies. We then generated summary statistics of BAFs for corn kernels and stover from the remaining studies.

2.3.2 | Cattle pharmacokinetic model

Most existing models assume a linear transfer (empirical toxicokinetic model) approach using transfer factors from feed to cattle (Maine Center for Disease Control & Prevention, 2020). The linear (static) model accurately predicted measured serum and muscle PFAS concentrations in cattle, provided the model was applied when intake rates remained constant over months (B. Astmann, 2024).

However, caution must be exercised when using these models in dynamic environments characterized by fluctuating intake, depuration, and accumulation periods, as averaging exposure over variable periods may lead to inaccuracies. Alternatively, physiologically based pharmacokinetic (PBPK) models are used to incorporate dynamic processes across organ compartments (German Federal Institute for Risk Assessment [BfR], 2024; Chou et al., 2023; Mikkonen et al., 2023; Savvateeva et al., 2020).

We adapted an existing one-compartment PBPK model that applies empirical serum-to-tissue ratios to predict PFOS concentrations in cattle serum, meat, liver, and kidney (Mikkonen et al., 2023). We included changes in body weight due to growth and dry matter intake (DMI) over a 2-year exposure period. We excluded other sources of variability in consumption rates, including interindividual and climatic differences. The daily PFOS exposure for cattle was quantified using the following Equation (1):

$$D = \left(C_{\text{feed}} \times I_{\text{feed}} \times B_{\text{feed}}\right) + \left(C_{\text{soil}} \times I_{\text{soil}} \times B_{\text{soil}}\right), \quad (1)$$

where D is the daily exposure to PFOS for cattle (ng/day), C is the concentration of PFOS in stover or soil (ng/g), I is the daily ingestion of stover or soil for cattle (g/day), and B is the average bioavailability of stover, soil, or water in the gut.

The model assumed a 2-year exposure period (730 days), capturing the typical growth cycle from calf to slaughter age. We modeled a hypothetical worst-case scenario in which beef cattle continuously and exclusively consumed corn stover (100%) from the field after harvest. This conservative assumption exceeds typical corn stover inclusion rates (30%–50%) in cattle diets (Chapple et al., 2015). Incidental soil ingestion (10%) occurs while grazing, and residual soil adhered to unwashed stover (Sun et al., 2022). Soil PFOS concentrations were measured directly in biosolid-impacted soils, and stover PFOS concentrations were measured and derived using literature-based BAFs. Assumptions were taken from Mikkonen et al. (2023), such as the growth rate constant of 0.0115 (Brody, 1945), an allometric scalar for growth of 0.27 (Taylor, 1968), and a mature cattle weight of 640 kg (Jenkins et al., 1993). DMI calculations used a relative size scalar of 0.025 and a quadratic constant of 1.7 (Mikkonen et al., 2023). Additional factors, such as a relative condition of 1 and a body condition factor of 1.5, ensured physiological relevance to agricultural practices (Mikkonen et al., 2023). Bioavailability assumptions for PFOS were derived from literature values, average of 50% and 90% for feed (EFSA CONTAM Panel et al., 2020), and 10%–30% for soil (Juhasz et al., 2025).

Our model uses a simplified assumption of continuous exposure for 2 years and distributes PFOS solely among the blood, liver, kidney, and meat without transgenerational transfer or additional exposure sources at an initial age of day 0 for young cattle (calf). The rate of change in the amount of PFOS in the body (AC) is given by Equation (2):

$$\frac{dAC}{dt} = \text{Dose} - AC \times \frac{\ln(2)}{\text{DT50}_t}, \quad (2)$$

where C is the amount of PFOS in the body (μg), Dose is the daily intake of PFOS (μg/day), and DT50_t is the time-dependent elimination half-life (days).

The elimination half-life ($DT50_t$) was adjusted for body weight using the following Equation (3):

$$\text{DT50}_t = \text{DT50} \times (\text{BW}/\text{MW})^{0.25}, \quad (3)$$

where DT50 is the elimination half-life (74.1 days; van Asselt et al., 2013), BW is the current body weight (kg), and MW is the mature weight (kg).

The corresponding serum concentration (C_{serum}) was calculated using the following Equation (4):

$$C_{\text{serum}} = AC/\text{Vd}, \quad (4)$$

where Vd is the volume of distribution (L), scaled with body weight (Vd = 0.085 × BW; Kowalczyk et al., 2013).

The tissue concentrations will be predicted by multiplying the C_{serum} by partition coefficients for meat (0.08), liver (1.44), and kidney (0.5), respectively (Mikkonen et al., 2023). A steady state was not achieved due to continuous PFOS exposure and its long elimination half-life (~74 days in non-lactating cattle), which prevented equilibrium within the 730-day model. To ensure model validation, we applied identical exposure and body weight assumptions. The cattle consumed PFOS-contaminated stover of 16.3 ng/g for 730 days to check the consistency of our results with ConTrans predictions (based on Mikkonen et al. [2023]). We compared the results from our model to those of an online model, ConTrans (based on Mikkonen et al. [2023]).

2.3.3 | Human exposure modeling

For this screening assessment, we selected a hypothetical scenario representing the worst-case scenario of elevated exposure conditions. The scenario assumes a family exclusively consumes corn and cattle from the impacted field. Dietary exposure was assessed for children (ages 1–2) and adults (ages 20–49) using median and 95th percentile body weight-adjusted consumption rates of corn grain, grain products, beef, and liver. These values were derived from the eaters-only population data in the National Health and Nutrition Examination Survey and reported in the EPA Exposure Factors Handbooks (US EPA, 2018).

To estimate daily PFOS intake from diet, we applied two exposure approaches: a central tendency (CT), which used

median PFOS concentrations and median consumption rates, representing average exposure conditions, and an upper bound (UB) approach based on the highest measured PFOS concentrations and 95th percentile consumption rates, representing a conservative, high-exposure scenario. Equation (5) was used to calculate the CT and UB estimated dietary intake (EDI, ng/kg-bw/day) from corn and cattle tissues (meat and liver) (Abunada et al., 2020; Brambilla et al., 2015; Costello & Lee, 2020; Felizeter et al., 2021; Ghisi et al., 2019; Lesmeister et al., 2021; Sinclair et al., 2020; Stahl et al., 2013; Surma et al., 2022; W. Wang et al., 2020; Xu et al., 2022).

$$\text{EDI (ng/kg-bw/day)} = \frac{C \times (BW - CR) \times (ED \times EF)}{AT}, \quad (5)$$

where EDI is the estimated dietary intake (dose [ng/kg-bw/day]), C is the concentration of PFAS in corn or cattle tissue (ng/g-ww), BW-CR is the body weight adjusted consumption rate (g/kg/day; US EPA, 2018), EF is the exposure frequency (days/years); assumed to be 365 days/year for daily exposure, ED is the exposure duration (years); a duration of 1 year was used, and AT is the averaging time (days); 365 days to estimate annual exposure.

We calculated the CT and UB hazard quotient (HQ) to assess the potential risk to human health using Equation (6), where a value >1 indicates that the exposure exceeds the reference dose (RfD) (US EPA, 2024). Further, a hazard index is calculated by summing the HQs from two sources in the case of PFOS.

$$HQ = \frac{EDI}{RfD}, \quad (6)$$

where HQ is the hazard quotient (unitless), EDI is the estimated daily intake (ng/kg-bw/day), and RfD is the reference dose (ng/kg-bw/day).

We obtained values from the United States Environmental Protection Agency (US EPA) for the RfD for chronic non-cancer effects. As of 2024, the US EPA RfD for PFOS is 0.1 ng/kg bw-day (US EPA, 2024).

3 | RESULTS

3.1 | PFAS measured in soil, corn stover, and kernels

Soil had detectable concentrations of 26 of the 45 investigated PFAS and was predominant for PFOS, which was detected in all samples (Figure 1; Table S5). Maximum and GM concentrations of PFOS (190 and 178 ng/g-dw) were higher for field composites compared to plot samples (110 and 97 ng/g-dw). Plot 4 had notably higher concentrations of 7:3 fluorotelomer carboxylic acid (FTCA; 10 ng/g-dw) and 5:3 FTCA (14 ng/g-dw) in soil compared to the other plots (<3 ng/g-dw). Other PFAS were detected in soil at concentrations <2 ng/g-dw.

Corn stover from the seven plots had detectable concentrations of six of the 45 investigated PFAS (Table S6). Stover was also predominant for PFOS (GM = 15.60 ng/g-dw), followed by PFBA, PFPrS, perfluoropropane sulfonate (PFPeA), and perfluorobutane sulfonate (PFBS; <1 ng/g-dw) (Table S6). The highest detection frequencies were observed for PFOS (100%), PFBS (100%), and PFBA (100%), followed by PFPrS (86%).

Corn kernels from composites and seven individual plots did not have detectable concentrations of any of the 45 investigated PFAS (RL < 0.05 ng/g).

3.2 | PFAS bioaccumulation factors

Median BAFs from soil to corn stover, calculated from our field study, were highest for PFBA (7), with decreasing values for PFBS (4), PFPeA (2), PFOA (0.3), and PFOS (0.2) (Figure 2; Table S7). We were unable to calculate BAFs for corn kernels as no PFAS were detected in those samples.

We identified five studies investigating PFAS uptake by corn stover from the literature (Blaine et al., 2013; Krippner et al., 2015; Navarro et al., 2017; Wen et al., 2016; H. Zhang et al., 2018) and excluded one located near a PFOS manufacturing plant that used PFAS-containing water for irrigation (Liu et al., 2019). Median BAFs were highest for PFBA (15), followed by PFPeA (11), PFBS (2), PFOA (0.3), and PFOS (0.2) (Tables S8 and S9).

For corn kernels, we identified four studies from the scientific literature. Only one of these identified PFOS in kernels and derived a low BAF for PFOS (<0.0002) (Felizeter et al., 2021; Table S10). That study had elevated concentrations in soils (>600 ng/g) and low concentrations in kernels (<0.4 ng/g). Soil concentrations for other PFAS (Table S11) detected in kernels were lower (<350 ng/g). One study with low concentrations in soils and a moderate reporting limit (0.1 ng/g) reported no detectable levels of PFAS in kernels (Blaine et al., 2013).

3.3 | PFAS calculated in corn stover and kernels using literature-derived BAFs

We calculated concentrations of five PFAS in corn stover by applying the literature-derived BAFs to PFAS concentrations measured in soil from each plot of our field study. The median calculated concentration of PFOS in corn stover (16.3 ng/g-ww) was similar to the concentration we measured in samples from our field study (16.6 ng/g-ww) (Figure 3; Table S12). Calculated concentrations of PFBA, PFPeA, and PFOA (9,

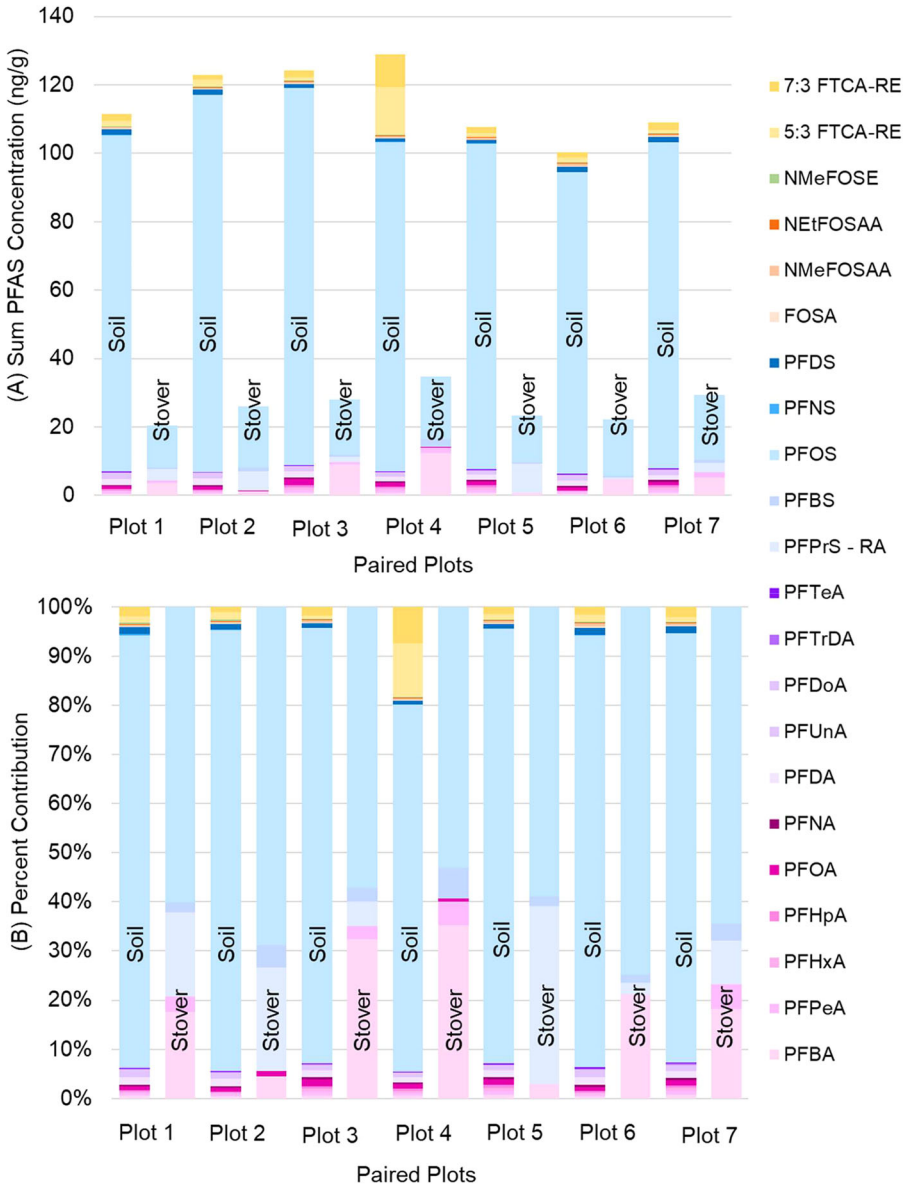


FIGURE 1 Concentrations (A) and percent contribution (B) of PFAS (ng/g) in soil and corn stover from seven paired plots.

4, and 0.33 ng/g-ww) were higher compared to those we measured (5, 1, and <0.19 ng/g-ww), and lower for PFBS (0.40 vs. 1 ng/g-ww).

3.4 | Cattle pharmacokinetic model

Predicted PFOS concentrations in cattle muscle (beef) were similar (30 ng/g vs. 29.5 ng/g) when using the median PFOS levels in stover from our field study and when using the median literature-based BAF (Figure 4). Median PFOS levels in cattle blood, liver, and kidney were also similar (375, 540, and 188 ng/g, respectively). However, the maximum PFOS concentrations in liver and beef were lower when using our stover data (420 and 34 ng/g) compared to when a range of BAFs was used (1783 and 99 ng/g). Despite these differences, PFOS levels in cattle tissues increased over time in all cases. Our results were also consistent with outputs from the ConTrans 2.0 model, with predicted beef concentrations falling within the model's median and lower values (Figure S2).

3.5 | Human estimated daily intake

Across all modeled exposure scenarios, PFOS EDIs were higher for children (ages 1–2) compared to adults (ages 20–49), based on body weight-adjusted consumption rates (Table S14). Consumption of the liver resulted in the highest CT EDIs for children and adults (811 and 216 ng/kg-bw/day; Figure 5; Table S15) and was threefold higher than



FIGURE 2 Comparison of soil to corn stover bioaccumulation factors (BAF) derived from previous studies (A) and calculated from our study samples (B). [Correction added on November 1, 2025, after first online publication: Figure 2 is updated; the y-axis label for the original figure was incorrectly labeled. The updated figure now indicate log10scale on the y-axis.]

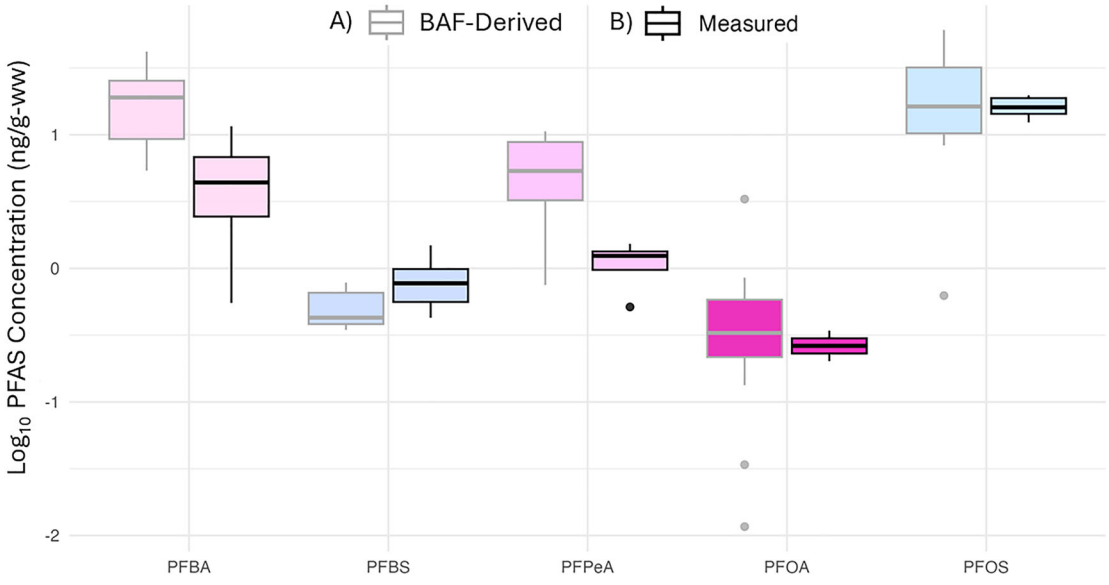


FIGURE 3 Comparison of bioaccumulation factor (BAF)-derived (A) and measured (B) concentrations of PFAS (ng/g-wet weight) in corn stover. [Correction added on November 1, 2025, after first online publication: Figure 3 is updated; the y-axis label for the original figure was incorrectly labeled. The updated figure now indicate log10scale on the y-axis.]

consumption of beef (muscle) (27 and 15 ng/kg-bw/day). EDIs for all scenarios were higher using PFOS predicted in beef from measured concentrations in our stover than those we calculated using literature-derived BAFs.

CT PFOS EDIs for children exceeded the 2024 EPA RfD (0.1 ng/kg-bw/day) and HQs from beef consumption were greater than 200 for children (HQ > 100 for adults) (Table S16). HQs were not calculated for corn kernel consumption due to the absence of detectable PFAS concentrations in kernels from our field study. Maximum EDIs did not exceed the PFOS RfD (HQ = 0.9) for children and adults based on kernel concentrations calculated using our field study soil concentrations and literature-derived BAFs.

Our model predicts that beef consumption can result in a PFOS CT EDI of 0.1 ng/kg-bw/day (US EPA, 2024) for children at lower concentrations in soil, stover, and beef (0.4, 0.06, and 0.1 ng/g-ww) compared to adults (0.8, 0.1, and 0.2 ng/g-ww, respectively) (Figure 6; Table S16). UB PFOS EDIs of 0.1 ng/kg-bw/day correspond to even lower concentrations in soil, stover, and beef.

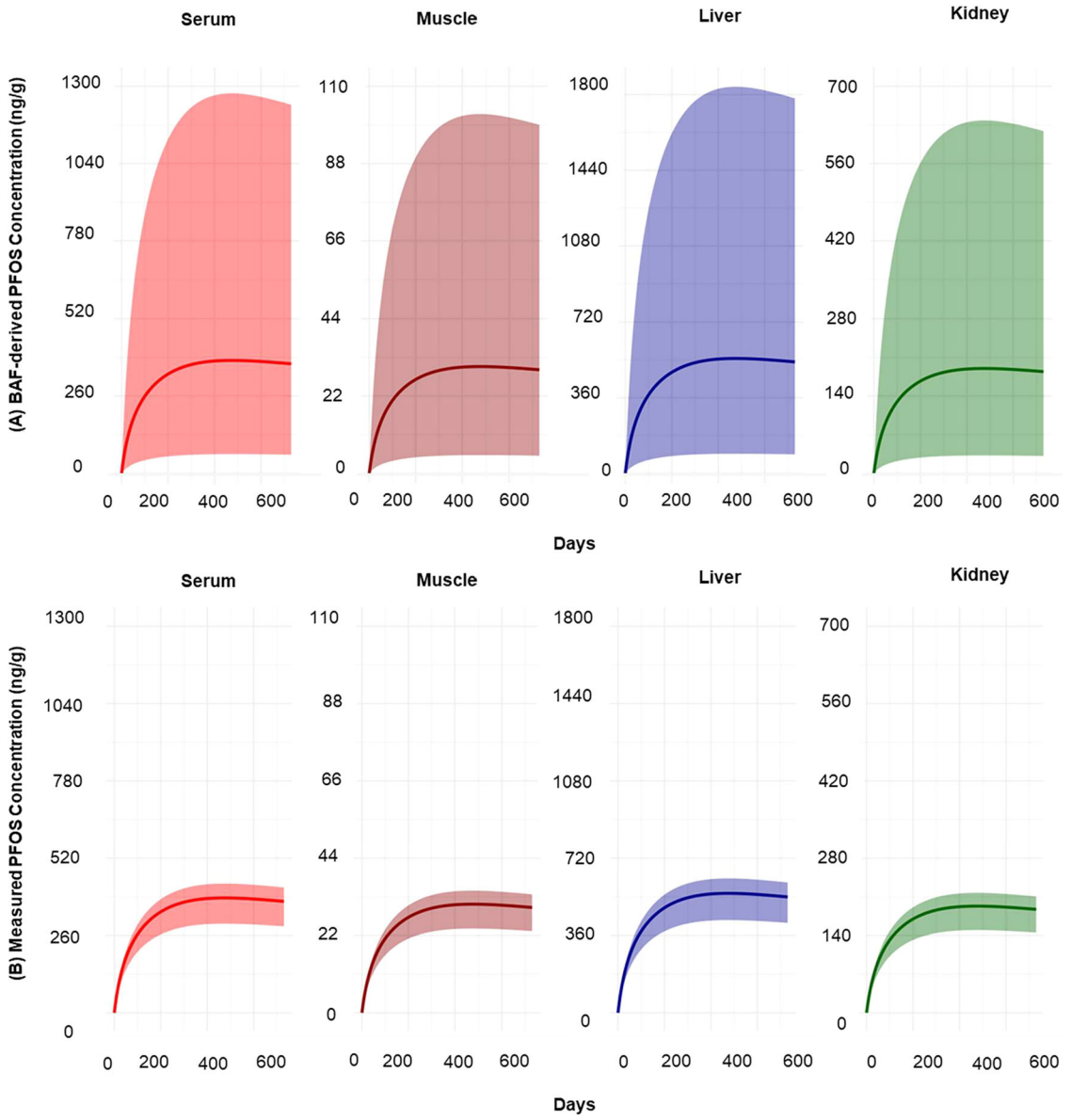


FIGURE 4 Modeled concentrations of perfluorooctane sulfonic acid (PFOS) in cattle serum, muscle, and liver from birth to slaughter (2 years) based on bioaccumulation factor (BAF)-derived (A) and measured (B) concentrations of per- and polyfluoroalkyl substances (PFAS) in corn stover. Each line indicates the median, and shading indicates the upper and lower confidence interval.

4 | DISCUSSION

4.1 | PFAS measured in soil, corn stover, and kernels

The soil had elevated PFOS concentrations (<100 ng/g-dw), yet only up to 19 ng/g-dw of PFOS was detected in corn stover, and no PFOS was detected in corn kernels (RL < 0.05 ng/g-ww). This is consistent with previous studies reporting higher PFAS accumulation in stover compared to kernels (Blaine et al., 2013; Krippner et al., 2015; Wen et al., 2016) (Tables S5 and S6). Stover also accumulated short-chain carboxyl PFAS, primarily PFBA and PFPeA (5–10 ng/g-ww), likely due to their greater mobility and translocation capacity (Lasters et al., 2024). Such PFAS have been frequently reported in livestock feed crops across the United States (Mroczko et al., 2022). Additionally, PFPrS-RA and PFBA were occasionally detected in stover but were absent from soil, likely reflecting contributions from irrigation water or atmospheric deposition (Figure 1; Table S6; Bao et al., 2020; Pepper et al., 2021; Pfotenhauer et al., 2022). Also, the quantification of PFBA is subject to variability due to the absence of a qualifying ion (Bangma et al., 2021). Kernels consistently showed no detectable PFAS, supporting the limited translocation of PFAS to seeds observed in previous controlled studies (Ghisi et al., 2019; Lesmeister et al., 2021).

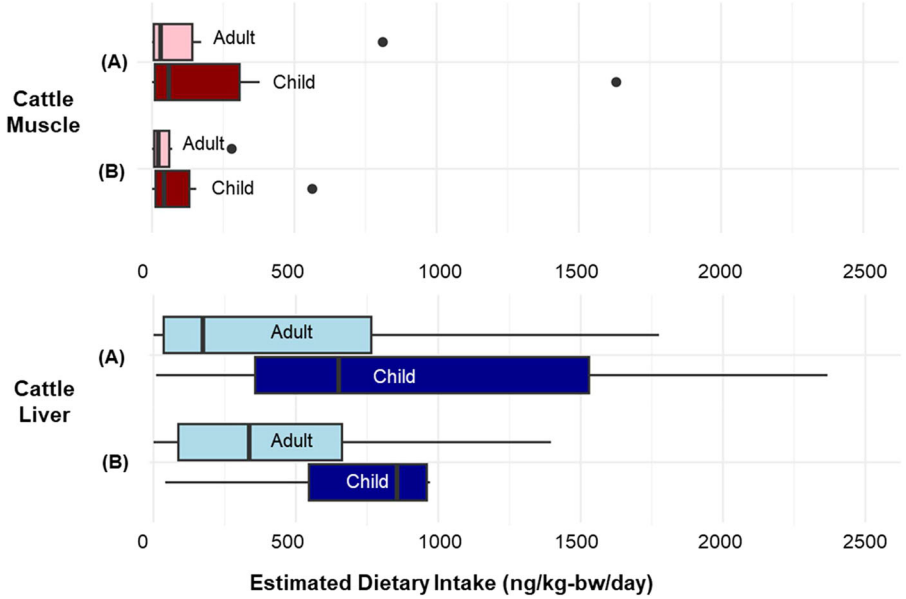


FIGURE 5 Estimated dietary intake (EDI) of PFOS for children (ages 1–2) and adults (ages 20 to 49) consuming cattle muscle and liver where estimated concentrations in these meat products are based on (A) bioaccumulation factor (BAF)-derived and (B) measured concentrations of perfluorooctane sulfonic acid (PFOS) in corn stover. Note that outliers from the 95th percentile consumption rate or maximum soil or stover PFOS concentrations.

4.2 | PFAS bioaccumulation factors

Measured BAFs for stover were lower than values reported in the literature, particularly for short-chain PFAS. Differences in soil concentrations and soil properties likely contributed to these differences (Ghisi et al., 2019; Lesmeister et al., 2021). Studies reporting higher BAFs often included soils with higher PFAS levels or soil properties such as organic carbon (<1.5%), soil type, and cation exchange capacity, which favor PFAS mobility and plant uptake (Blaine et al., 2013; Krippner et al., 2015; Navarro et al., 2017; Wen et al., 2016; H. Zhang et al., 2018). Additionally, environmental conditions such as rainfall and irrigation patterns may influence plant uptake but were not uniformly reported across studies (Adu et al., 2023).

BAFs for kernels reported in the literature were similarly low or not calculable due to non-detects, consistent with our findings. The other two studies that investigated PFOS did not detect it in kernels, although one had a high reporting limit (1 ng/g; Krippner et al., 2015) and the other had low concentrations in soil (<5 ng/g; Blaine et al., 2013).

Differences in analytical methods may also contribute to variability in BAF across the literature. Higher reporting limits in some studies may have precluded the detection of low-level PFAS in the stover and kernels. In our analysis, an interference peak near the PFOS retention time was observed in kernel composites (Figure S3), consistent with challenges noted by others when working with complex food matrices (Genualdi & deJager, 2024; Genualdi et al., 2022). Such interferences, if unresolved, may lead to false positives or inaccurate quantification. When applying BAFs for exposure and risk assessments, careful consideration of site-specific conditions, plant physiology, and analytical limitations is crucial to reduce uncertainties (Costello & Lee, 2020).

4.3 | Cattle pharmacokinetic model

Predicted PFOS concentrations in cattle muscle increased from 1.2 ng/g at Day 3 to 30 ng/g and 29.5 ng/g at Day 730 using measured and literature-derived BAFs, respectively. These values exceed the US EPA screening level of 4.1 ng/g (US EPA, 2016), Maine's actionable level of 3.4 ng/g (Maine Center for Disease Control & Prevention, 2020), and the European Commission's maximum contamination levels of 0.3 ng/g for beef (EUR-Lex, 2022). Our results are two to three times higher than the blood-based model developed by Johnston et al. (2023) from a depuration study in cattle near military sites. Concentrations are predicted 11 times higher than those predicted by B. A. Astmann et al. (2025), who applied a dynamic model accounting for seasonal depuration, lower PFOS (≤19.8 ng/g) in soil, on clean pasture, and lower soil ingestion rates (4% DMI). Also, the predicted PFOS in cattle muscle is five to seven times higher than a field study with cattle exposed to contaminated biosolids (<2150 ng/g, Klase et al., 2023). The higher concentrations predicted by our investigation are likely due to continuous exposure to contaminated stover and soil over time. Other field studies

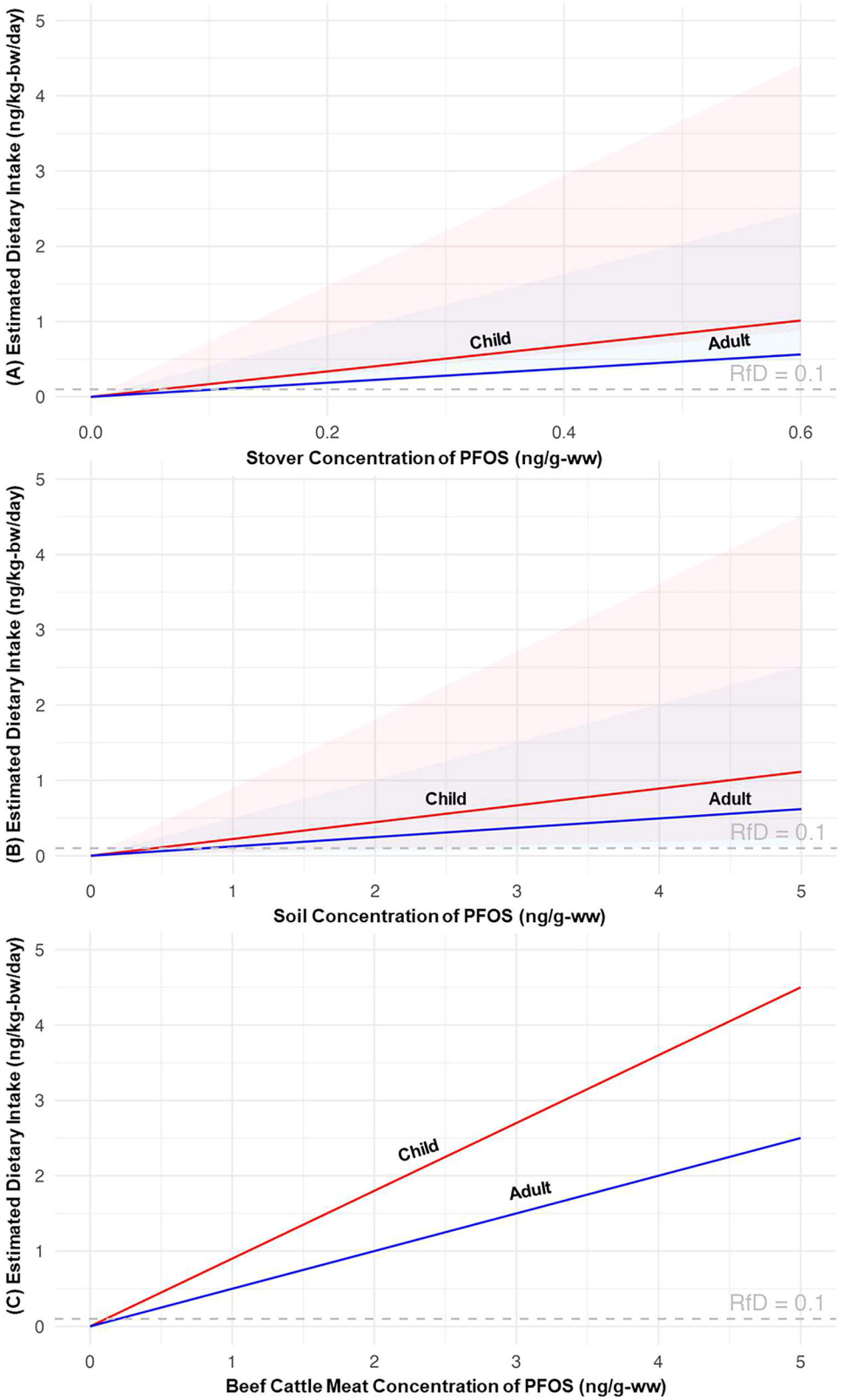


FIGURE 6 Estimated dietary intakes (EDI, ng/kg-bw/day) of perfluorooctane sulfonic acid (PFOS) for children (ages 1 to 2, the red line is median EDI) and adults (ages 20–49, the blue line is median EDI) from our model for the soil-stover-meat pathway across concentrations of PFOS in (A) stover, (B) soil, and (C) beef cattle meat. Shading indicates upper-bound EDIs for children (red) and adults (blue). The dashed gray line at 0.1 ng/kg-bw/day indicates the U.S. EPA's reference dose (RfD) for PFOS.

have found lower concentrations of PFOS in cattle meat, such as in the United States (8.1–14.2 ng/g; Lupton et al., 2025), Australia (5.8 ng/g; Drew et al., 2021), the Netherlands (0.09 ng/g; Noorlander et al., 2011), Italy (2.1 ng/g; Guerranti et al., 2013), other European countries (0.08 ng/g; Ullah et al., 2012), and China (4.4 ng/g; T. Zhang et al., 2010). The variability across studies is expected due to differences in contamination sources, agricultural practices, and environmental conditions (G. Wang et al., 2017).

PFOS concentrations were consistently higher in the liver than in the cattle muscle. Short-term predictions estimated liver PFOS at 21.6 ng/g by Day 3, increasing to over 540 ng/g (measured) and 530 ng/g (literature derived) at Day 730. Other studies have reported lower liver concentrations, typically for shorter exposure periods up to 77 ng/g (Guruge et al., 2008), up to 24 ng/g (Drew et al., 2021), and up to 200 ng/g (Lupton et al., 2025). The elevated liver concentrations are consistent with the liver's role as a key PFOS storage organ, due to its strong binding to liver-specific proteins, as well as slower elimination relative to muscle (Das et al., 2017; Pérez et al., 2013).

For this hypothetical scenario, immediate management strategies are necessary. Short-term measures include removing cattle from contaminated agricultural fields, changing feed sources, and implementing soil remediation (Mikkonen et al., 2023). Using the available half-life values for PFOS elimination in cattle tissues (Drew et al., 2021), it would take approximately 5.5 years for blood, 2.5 years for meat, 6 years for liver, and 6.5 years for kidney PFOS levels to go below 1 ng/g after removing exposure sources.

4.4 | Human estimated daily intake

Cattle meat and liver are known to take up PFOS and transfer it to humans via diet (EFSA CONTAM Panel et al., 2020). Exposures estimated using both measured and literature-derived BAF scenarios were elevated for cattle meat and liver, with the highest exposures observed in children (1–2 years old) due to their higher intake relative to body weight (US EPA, 2016). Cattle liver consistently contributed the highest exposures, followed by cattle meat, while corn kernels contributed negligible amounts. Intake rates were based on consumer-only data (Table S14). Although the cattle liver is consumed by fewer individuals and at lower amounts than muscle, its higher PFOS concentrations result in disproportionately higher EDIs (Figure 5; Table S15). This pattern is consistent with previous studies where cereal crops, including corn, contributed minimally to PFOS dietary exposure (<0.005 ng/kg-bw/day) (Chou et al., 2023; EFSA CONTAM Panel et al., 2020).

Adults consuming cattle meat from our predictions were two to three times higher than those reported in the Chinese Total Diet Study (Xiao et al., 2024) and Wang et al. (2017), ranging from 0.009 to 4 ng/kg-bw/day. A recent Polish farm study reported lower exposures up to 0.9 ng/kg-bw/day for adults (Mikolajczyk et al., 2024). Even for shorter durations, such as 120 days of meat consumption, our EDIs exposures are 8 ng/kg-bw/day for children and 5 ng/kg-bw/day for adults. Similarly, cattle liver exposures from our model were several magnitudes higher than liver EDIs reported by Wang et al. (2017, up to 3.3 ng/kg-bw/day).

Cereal crops like corn contribute little to PFOS exposure (EFSA CONTAM Panel et al., 2020). Other studies have found little to no exposure to PFOS in crops, including corn, grown in areas with PFAS contamination ranging from 0.001 to 0.005 ng/kg-bw/day (Chou et al., 2023).

HQs using the 2024 EPA RfD (0.1 ng/kg-bw/day) consistently indicated potential risk (>1) from both the literature-derived BAF and measured approach, assuming 100% of consumption comes from this contaminated supply. HQs values are several magnitudes higher, especially for children consuming cattle liver followed by meat, consistent with a few other studies (G. Wang et al., 2017). Most of the previous literature used a higher reference dose (such as an RfD of 20 ng/kg-bw/day) and found HQs < 1 in the cattle meat and liver (G. Wang et al., 2018). HQs could not be calculated for kernels due to non-detects, consistent with the minimal role of corn in dietary PFOS exposure. Our results suggest that consuming these cattle at the median or 95th percentile consumption rates for adults and children will likely lead to exceedances of the selected toxicity value, increasing the risk of health effects with long-term exposure.

Also, our soil screening levels derived from PFOS CT EDI of 0.1 ng/kg-bw/day (US EPA, 2024) are consistent with the recent US EPA-drafted guidelines for PFAS in sludge, suggesting that any soil with PFOS concentrations exceeding 1 ng/g poses a risk to communities consuming livestock and agricultural products from the impacted field (Figure S4; US EPA, 2025).

The strengths of our study include a literature review of BAFs based on soil and corn kernel and stover uptake data. Further, we used a published compartment-based cattle modeling approach to model the PFOS concentrations in cattle blood, meat, liver, and kidney and validated these models using online software. We used conservative exposure modeling assumptions, including younger age groups with higher body weight-adjusted consumption rates. We considered seasonal and yearly consumption of cattle and corn exclusively from the impacted field. The limitations include the observed root ball contamination, which decreased the number of stover samples tested from individual plots. We limited our predicted cattle meat and liver results to PFOS, and future evaluation should include a broader range of current-use and other short-chain PFAS. This model excludes additional PFOS sources, such as irrigation and runoff water contamination. Finally, the stover samples were analyzed by the contract lab only.

5 | CONCLUSIONS

Our results show a higher uptake of PFAS into stover compared to corn kernels, which is consistent with several prior studies. Differences in methods for PFAS testing, atmospheric deposition near an industrial source, and interferences during analysis may contribute to the wide range of BAFs identified in our literature review. Hence, the study highlights the need for caution when using literature-derived BAFs, as variability in environmental conditions, literature review, and analysis can lead to uncertainties in exposure and risk assessments.

We used a published compartment-based pharmacokinetic model consuming impacted corn stover and soil for 730 days. We found compartment (such as liver and meat) concentrations in the order of blood > liver > kidney > meat. Our results suggest that any soil with PFOS concentrations exceeding 1 ng/g poses a risk to communities consuming livestock and agricultural products from the impacted field. Children's exposure to predicted meat and liver consumption was significantly higher than that of adults due to body weight-adjusted intake rates. Our results suggest that consuming these cattle tissues at the median or higher consumption rates for adults and children will likely lead to exceedances of the selected toxicity value and increase the risk of health effects given long-term exposure. Finally, we found that directly measuring PFOS in crops can substantially reduce variability in modeling and the human exposure assessment process.

AUTHOR CONTRIBUTIONS

Ankita Bhattacharya: Conceptualization; data curation; formal analysis; investigation; methodology; project administration; resources; software; validation; visualization; writing—original draft. **Jiahe Tian**: Formal analysis. **Sarah Choyke**: Formal analysis; writing—review and editing. **Juliane Brown**: Writing—review and editing. **PingPing Meng**: Formal analysis; writing—review and editing. **Geoff Rhodes**: Conceptualization; formal analysis; writing—review and editing. **Andrew Patterson**: Formal analysis. **Detlef R. U. Knappe**: Formal analysis; writing—review and editing. **Hui Li**: Formal analysis; writing—review and editing. **Christopher P. Higgins**: Formal analysis; writing—review and editing. **Courtney Carignan**: Conceptualization; funding acquisition; project administration; supervision; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

Courtney Carignan and Christopher P. Higgins have served as expert consultants in litigation related to PFAS-contaminated drinking water. All other authors declare no conflicts of interest.

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